Event-Timing Priors: Temporal Point Process Likelihoods Correct the Long-Horizon Extreme-Event Statistics of Generative Surrogates

Gunner Howe
gunnerlevihowe@gmail.com

July 2026

Abstract

Generative surrogates of chaotic dynamical systems — including modern diffusion-based weather and climate emulators — are routinely *evaluated* on extreme events but never *trained* on their timing. We show that this omission is consequential and correctable. Working with a two-scale Lorenz-96 testbed observed under realistic conditions (coarse emulator timestep, observation noise), we find that a conditional flow-matching surrogate reproduces marginal statistics well while systematically corrupting the *temporal* statistics of threshold-crossing extreme events in long rollouts: event rates inflate, inter-event-time distributions distort, and return periods drift. We introduce an *event-timing prior*: a neural temporal point process (TPP) fit to the event timings of a curated catalog, whose likelihood — evaluated on soft, differentiable events extracted from short surrogate rollouts — serves as an auxiliary fine-tuning signal. Naive likelihood maximization is mode-seeking and collapses the event rate; we instead use a likelihood-ratio objective against a co-trained "self-TPP" (the point-process analog of variational score distillation), whose gradient vanishes exactly when the surrogate's event process matches the catalog's. Across seeds, TPP-aux fine-tuning recovers inter-event-time distributions, burstiness, and return-period curves that invariant-measure matching [6] provably leaves unconstrained, while a rate-matched Poisson control shows the gains require the prior's temporal *structure*, not merely its rate. Event-timing structure is thus a distinct, directly controllable axis of surrogate quality.

1 Introduction

Data-driven surrogates now emulate weather and climate at a fraction of the cost of numerical models, and generative (diffusion-family) surrogates such as GenCast [18] explicitly target probabilistic skill. A striking asymmetry runs through this literature: extreme events — heat waves, cyclones, blocking episodes — dominate the *evaluation* of these models [8, 18, 24], yet no training objective in use says anything about *when* extremes occur. Standard objectives constrain pointwise forecast error or, at best, time-aggregated distributional statistics: invariant measures [6, 20], spectra [4], or adversarial OT statistics [15]. All of these are invariant to permuting time — they cannot distinguish a process whose extremes arrive quasi-periodically from one whose extremes arrive in bursts, so long as the marginals agree.

The temporal law of extremes is precisely what many downstream users care about: return periods drive design codes; clustering of exceedances drives compound risk; inter-event times drive recovery windows. And it is a quantity for which independent, high-quality data often exists even when state observations are noisy: curated event catalogs (hurricane best tracks, earthquake catalogs, flood records) are compiled and quality-controlled separately from gridded reanalyses.

Contributions.

1. We demonstrate a clean dissociation: under realistic training conditions (coarse timestep, noisy observations), a strong generative surrogate's long-horizon rollouts corrupt event-timing statistics in ways that invariant-measure/marginal-statistics training does not and cannot fix (§5).

2. We introduce TPP-aux fine-tuning: a neural TPP fit to catalog event timings acts as a differentiable prior on the timing of soft events extracted from short surrogate rollouts (§3). We identify and solve the mode-seeking pathology of naive TPP-likelihood maximization via a likelihood-ratio (self-TPP) objective, the point-process analog of variational score distillation [12, 23].

3. Controls that isolate the mechanism: a rate-matched *Poissonized* TPP prior (same loss form and strength, no temporal structure) fails to deliver the gains; the invariant-measure objective of Jiang et al. [6] (given equally privileged clean statistics) fixes marginals but not timing; pushforward-style stabilization [3] fixes neither.

2 Related work

Long-time statistics of neural surrogates. Jiang et al. [6] train neural operators to preserve invariant measures of chaotic attractors via an optimal-transport (Sinkhorn) loss on physics-informed summary statistics, or a contrastive feature loss; DySLIM [20] regularizes toward the invariant measure; spectral losses match frequency content [4]; adversarial OT regularization matches trajectory statistics [15]. These objectives are functions of the *time-marginal* (or time-averaged) law of the process and are therefore blind to the ordering and spacing of rare events; we use Jiang et al. [6] as our primary control. Rollout stabilization — the pushforward trick [3], noise injection and refinement [10, 17], Jacobian regularization [16] — targets stability, an orthogonal axis.

Extremes and generative emulators. Stamatelopoulos and Sapsis [22] study when diffusion models extrapolate amplitude tails; Manshausen et al. [13] extract extreme-event likelihoods from a trained diffusion emulator at inference time. Flagship emulators evaluate extremes diagnostically [8, 18, 24]. None of these place event *timing* in the training signal. Raeth and Ludwig [19] augment generative forecasters with differentiable decision losses — structurally the closest training-objective precedent — but target decision cost, not temporal event structure, and no dynamical rollout is involved.

Neural temporal point processes. Neural TPPs model conditional intensities of event sequences [14] or inter-event densities directly [21]; see [1] for benchmarking. To our knowledge, TPP likelihoods have not previously been used as training objectives for dynamical-system surrogates.

3 Method

3.1 Setup: surrogates, events, catalogs

Let $\mathbf{x}_t \in \mathbb{R}^K$ be the observed (slow) state of a chaotic system sampled at interval Δt. A generative surrogate models $p_\theta(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$; long rollouts compose samples autoregressively. We train on noisy observations $\tilde{\mathbf{x}}_t = \mathbf{x}_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, r^2\sigma_x^2 I)$, reflecting the practical setting where state data comes from noisy analyses [6].

Events are threshold upcrossings of a scalar site observable $s_k(t)$ (here $s_k = -x_k$, deep troughs; threshold $u =$ the pooled 95th percentile): an event occurs at step t of stream k iff $s_k(t) > u$ and $s_k(t-1) \le u$. By rotational symmetry the K site streams are exchangeable. We assume access to a *catalog*: the event timings extracted from the clean historical record (in practice, curated event databases; in our testbed, events of the clean trajectories underlying the noisy training data). Crucially, the catalog contains *event times only* — orders of magnitude less information than the full state record.

3.2 The event-timing prior: a recurrent-intensity TPP

We fit a discrete-time recurrent-intensity TPP $p_{\phi^\star}$ to catalog streams: a GRU consumes the binary event sequence $w_{1:T}$ of one stream and emits a per-step conditional intensity $\lambda_t = \mathrm{softplus}(g(h_{t-1}))$ (events per unit time, depending only on events strictly before t), with log-likelihood

$$\mathcal{L}(w;\lambda) \;=\; \sum_t \big[\, w_t \log \lambda_t \;-\; \lambda_t \, \Delta t \,\big], \tag{1}$$

the standard grid discretization of the continuous point-process likelihood [5, 14], exact as $\Delta t \to 0$. Two properties matter. First, (1) is *linear in* w_t, so relaxed (soft) event indicators substitute cleanly — this is what makes the prior differentiable end-to-end. Second, the same fitted model evaluates any candidate event stream, which is how it becomes a training signal.

3.3 Soft events from differentiable rollouts

From a rollout $\hat{\mathbf{x}}_{0:T}$ we extract soft upcrossing weights

$$w_t \;=\; \sigma\big((s_t - u)/\tau\big)\, \sigma\big((u - s_{t-1})/\tau\big) \in [0,1], \tag{2}$$

with temperature τ calibrated so the soft event rate matches the hard rate on held-out data (bias $< 5\%$ at our τ). The surrogate's few-step deterministic sampler makes each rollout step differentiable given its noise draw, and gradients through the chaotic dynamics are truncated to windows of T_w steps (states are recorded in a gradient-free rollout, and all windows are re-forwarded with gradients in parallel from their recorded start states — identical values and gradient semantics to truncated BPTT, at a fraction of the sequential cost). The TPP consumes the full stream: each rollout is prefixed with the catalog's event history preceding the initial condition, so intensities are history-calibrated where the likelihood begins to count.

3.4 Mode-seeking, and the likelihood-ratio fix

The obvious auxiliary loss, $-\mathbb{E}[\mathcal{L}(w^{\mathrm{soft}};\lambda_{\phi^\star})]$, is a density, not a fit measure: under a fitted TPP the *empty* sequence often has higher density than typical sequences, so maximizing catalog-TPP likelihood of rollout events drives the surrogate toward the TPP's mode — suppressing events rather than matching their law. We observe exactly this collapse empirically (§6). The fix mirrors variational score distillation [12, 23]: maintain a small *self-TPP* p_ψ, refit continually by maximum likelihood on the surrogate's own (hard, detached) rollout events, and optimize the ratio

$$\mathcal{L}_{\mathrm{aux}}(\theta) \;=\; -\,\mathbb{E}_{\mathrm{rollouts}}\big[\, \mathcal{L}(w^{\mathrm{soft}};\lambda_{\phi^\star}) - \mathcal{L}(w^{\mathrm{soft}};\lambda_\psi)\,\big]. \tag{3}$$

In expectation this is the reverse KL between the surrogate's event process and the catalog process (up to the self-TPP's fit error): its gradient vanishes exactly when the two laws agree, and the self-TPP term cancels the uniform "fewer events are more likely" pressure. Fine-tuning minimizes $\mathcal{L}_{\mathrm{FM}} + \lambda_{\mathrm{aux}}\mathcal{L}_{\mathrm{aux}}$, keeping the base flow-matching loss as an anchor on ordinary skill.

3.5 Conditions

All conditions fine-tune the *same* per-seed base checkpoint for the same number of steps; they differ only in the auxiliary objective: **base** (flow-matching only, compute-matched); **TPP-aux** (ours, Eq. 3 with the catalog TPP); **Poisson-aux** (identical, but the prior TPP is fit to Poissonized catalogs — same rate, exponential inter-event times; the content-specificity control); **invariant-stats** (the OT objective of Jiang et al. [6]: debiased Sinkhorn divergence between rollout and clean-climatology clouds of their physics-informed summary statistics $\{\dot{u}_i, (u_{i+1}-u_{i-2})u_{i-1}, u_i\}$ — given the same clean-reference privilege as the catalog); **pushforward** [3]; **deterministic** (MSE-trained autoregressive baseline).

4 Experimental setup

System. Two-scale Lorenz-96 [11, 25]: K=40 slow sites, J=10 fast variables per site (F=10, h=1, b=c=10), RK4 at dt=0.002, observed at $\Delta t = 0.2$ MTU (coarse emulator step). Training: 64 trajectories $\times$ 160 MTU with observation noise r=0.3; evaluation references from clean held-out runs (26,600 MTU total).

Ground-truth event structure. Trough events are strongly sub-Poisson (quasi-periodic recurrence): IET CV 0.78, Fano factors 0.6–0.7 across windows; near-renewal (lag-1 IET correlation ≈ 0). Aggregate (spatial-max) events are super-Poisson (CV 1.15) — used as a held-out transfer diagnostic.

Models. Surrogate: conditional flow matching [9] on normalized state residuals; 1D circular CNN (4 residual blocks, width 96, FiLM noise-level conditioning, ∼0.66M parameters); 6-step Euler sampler. TPP: single-layer GRU, hidden 64 (∼40k parameters). Full hyperparameters in appendix.

Metrics. Event timing: pooled IET distribution (two-sample KS, W_1), event rate ratio, Fano-factor curve error, return-period curves at catalog quantiles $q \in \{0.9, \ldots, 0.995\}$, catalog-TPP held-out log-likelihood of rollout events. Ordinary skill: pooled state W_1, PSD log-distance, ACF error, short-horizon CRPS and ensemble spread. Stability: diverged-rollout fraction (censored and reported). ≥ 5 confirmatory seeds, paired by shared base checkpoint; the tuning seed is excluded.

5 Results

5.1 Lorenz-96: the event-timing deficit and its repair

The base surrogate corrupts event timing. Trained on noisy observations, the base surrogate's long rollouts (64 $\times$ 800 MTU) inflate the event rate by 10.0% $\pm$ 1.2%, distort the inter-event-time distribution (IET KS 0.065 ± 0.007, W_1 0.41 ± 0.04 MTU), and shift return periods (mean $|\log \mathrm{RP}_{\mathrm{model}}/\mathrm{RP}_{\mathrm{true}}| = 0.105 \pm 0.009$ across catalog quantiles 0.90–0.995). An identically-budgeted continuation of flow-matching training (**base** condition) leaves all of these unchanged — the deficit is persistent, not undertraining.

TPP-aux repairs it without costing skill. Fine-tuning with the event-timing prior recovers the event rate (1.005 ± 0.035; paired t-test vs. base $p = 0.006$), the IET distribution (KS 0.028 ± 0.004, $p = 2 \times 10^{-4}$; W_1 0.14 ± 0.07, $p = 0.001$), and return periods (0.055 ± 0.033, $p = 0.038$). The catalog-TPP log-likelihood of rollout events rises to -0.0992 ± 0.0012 — statistically indistinguishable from the value the catalog TPP assigns to *real* held-out event sequences (-0.0994): the fine-tuned surrogate's events are exactly as typical under the prior as truth itself, neither less (deficit) nor more (mode-seeking). Short-horizon CRPS changes by +0.4% ($p = 0.16$); no rollout diverged in any condition (Table 1).

Table 1: Lorenz-96 main results: mean ± s.d. over confirmatory seeds (shared base checkpoint per seed; identical fine-tuning budgets). Bold = best among fine-tuned generative conditions. IET = inter-event time; RP = return period; TPP-LL = catalog-TPP log-likelihood of rollout events (ground-truth ceiling −0.0994); CRPS at lead 20 steps.

condition	IET KS	IET W_1	rate ratio	RP log-err	TPP-LL	state W_1	CRPS@20
Base (FM only)	0.065 ± 0.007	0.414 ± 0.044	1.100 ± 0.012	0.105 ± 0.009	-0.1018 ± 0.0001	0.094 ± 0.014	$\mathbf{1.735 \pm 0.002}$
TPP-aux (ours)	$\mathbf{0.028 \pm 0.004}$	$\mathbf{0.142 \pm 0.068}$	$\mathbf{1.005 \pm 0.035}$	0.055 ± 0.033	-0.0992 ± 0.0012	0.066 ± 0.045	1.743 ± 0.008
Poisson-TPP-aux	0.095 ± 0.018	1.054 ± 0.215	0.809 ± 0.032	0.229 ± 0.075	$\mathbf{-0.0946 \pm 0.0013}$	0.153 ± 0.032	1.769 ± 0.014
Invariant-stats	0.030 ± 0.005	0.211 ± 0.087	0.965 ± 0.034	0.117 ± 0.043	-0.0971 ± 0.0015	$\mathbf{0.048 \pm 0.019}$	1.755 ± 0.019
Pushforward	0.047 ± 0.003	0.142 ± 0.023	1.029 ± 0.007	$\mathbf{0.036 \pm 0.006}$	-0.1011 ± 0.0002	0.076 ± 0.005	1.767 ± 0.003
Deterministic AR	0.173 ± 0.082	1.021 ± 0.453	1.310 ± 0.197	0.454 ± 0.246	-0.1030 ± 0.0117	0.473 ± 0.151	3.225 ± 0.023

Structure, not rate: the Poisson control. The Poissonized prior — identical loss form, identical strength, identical rate information, exponential inter-event times — does not merely fail; it actively damages the surrogate (IET W_1 1.05 ± 0.21, $p = 0.004$; rate ratio 0.81 ± 0.03, $p = 3\times10^{-5}$; Fano error $+0.058$, $p = 7\times10^{-5}$). Its rollouts' hazard function is driven toward a constant (Fig. 2), erasing the quasi-periodic recurrence structure the dynamics should carry: optimizing toward a structureless event law distorts the dynamics that produce events. The gains of TPP-aux are therefore attributable to the *temporal structure* of the prior, not to rate matching or to the loss mechanics.

What the timing prior transfers. Ground-truth trough events are quasi-periodic: the IET distribution is multimodal at harmonics of the wave-recurrence period, and the conditional hazard oscillates (Fig. 2, black). The base surrogate over-amplifies the hazard oscillation; TPP-aux pulls it onto the ground-truth curve; the Poisson prior flattens it. On the IET quantile–quantile plot (Fig. 1), TPP-aux is the only condition tracking the diagonal across the full range.

Comparison with time-marginal objectives. The invariant-statistics control [6], given clean summary-statistic clouds of equal privilege, achieves the best state marginals and spectra (as designed) and — in this noise-dominated regime, where the base's timing errors are largely inherited from amplitude corruption — also improves IET-level statistics. But it does not improve return periods (0.117 ± 0.043 vs. base 0.105, $p = 0.61$), significantly worsens burstiness (Fano error $p = 0.023$), undershoots the event rate (0.965), and its rollout events are *more* typical than truth under the catalog TPP (−0.0971, overshooting the ceiling). The pushforward control improves rate-side statistics (return periods 0.036 ± 0.006) — consistent with its denoising effect on rollouts — but leaves the conditional timing structure untouched (catalog-TPP LL −0.1011, at base level) and significantly degrades spectra ($p = 10^{-5}$) and CRPS ($p = 5\times10^{-5}$). Only TPP-aux exhibits the full pattern: conditional timing at the ground-truth ceiling, marginals improved, and ordinary skill preserved. The deterministic AR baseline, for reference, is far worse on every axis (CRPS 3.23 vs. 1.74; state W_1 0.47), confirming that generative surrogates are the right object of study.

5.2 Kuramoto–Sivashinsky: a purely temporal deficit

On KS ($L = 22$, $N = 64$, $\Delta t = 1$ tu, events = upcrossings of u at its 95th percentile; *clustered*, IET CV 1.58 — the opposite deviation from Poisson to Lorenz-96), the noisy-observation base surrogate exhibits a *purely temporal* deficit: its rollout state marginal is nearly perfect (state $W_1 = 0.037$) while the event process is severely corrupted — 2.5× the true event rate, IET KS 0.59, return-period log-error 1.14. Here there is nothing for a time-marginal objective to fix by construction; the corruption lives entirely in temporal structure.

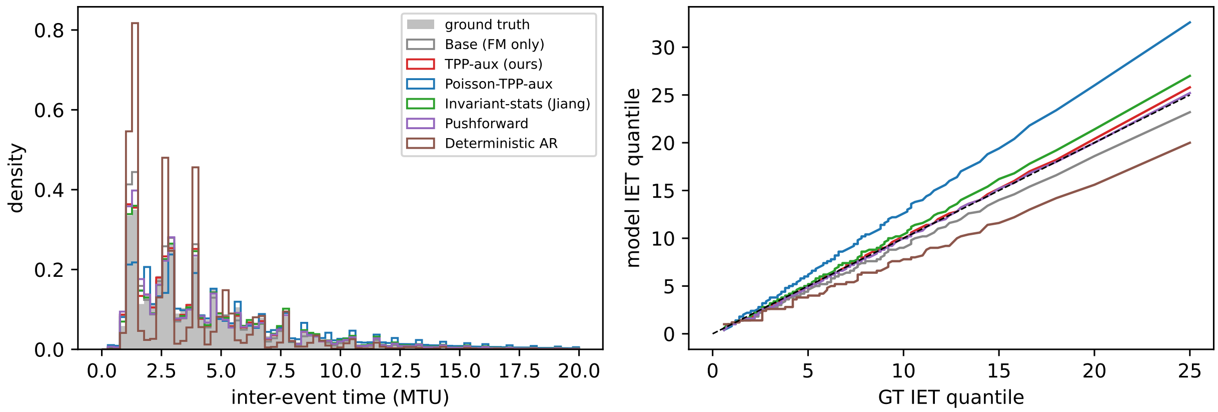

Figure 1: Inter-event-time statistics, Lorenz-96 (pooled over 5 seeds). Left: IET densities; ground truth (shaded) is multimodal at wave-recurrence harmonics. Right: quantile–quantile against ground truth; TPP-aux (red) tracks the diagonal; the Poissonized prior (blue) severely distorts the distribution.

6 Ablations

Naive likelihood collapses. Replacing the ratio objective with plain likelihood maximization ($-\mathbb{E}[\log p_{\phi^\star}]$, no self-TPP) suppresses 99.75% of events (rate ratio 0.0025) while *increasing* the catalog-TPP log-likelihood of rollouts to -0.055, far above the ground-truth ceiling of -0.0994: near-empty sequences really are the density mode of a fitted TPP, and a surrogate optimized toward that mode stops producing extremes at all. The self-TPP ratio term is not a refinement but a requirement.

Which structure matters: the shuffled control. A prior fit to interval-*shuffled* catalogs (IET marginal preserved, serial correlations destroyed) performs on par with the full prior (IET KS 0.022, rate 0.995, seed 0), consistent with the near-renewal character of the ground-truth event process (lag-1 IET correlation ≈ 0). Together with the Poisson control this brackets the operative content of the prior: the *shape of the inter-event-time distribution* — destroyed by Poissonization, preserved by shuffling — is what repairs the surrogate.

Soft-event temperature. At our τ the soft event rate is biased +4.9% with 80% of soft mass on true hard-event steps; doubling τ inflates the bias to 17.5% (appendix).

7 Discussion and limitations

Why fine-tuning with a tiny prior works. The catalog TPP has ~40k parameters and sees only event times — a few thousand numbers — yet repairs a surrogate trained on six orders of magnitude more state data. Event timing is a low-dimensional, data-efficient summary of exactly the aspect of the dynamics that ordinary training leaves unconstrained under observation noise. This information asymmetry mirrors practice: curated event catalogs (best tracks, earthquake catalogs, flood records) are compiled independently of — and are far cleaner than — gridded state estimates.

Limitations. (i) Our systems are canonical chaotic testbeds, not full emulator stacks; scale remains to be demonstrated. (ii) The catalog is assumed accurate; catalog noise/censoring would enter the prior. (iii) The TPP likelihood is grid-discretized at the emulator step (exact only as $\Delta t \to 0$). (iv) Truncated rollout gradients bias the aux gradient; DDPO-style score-function

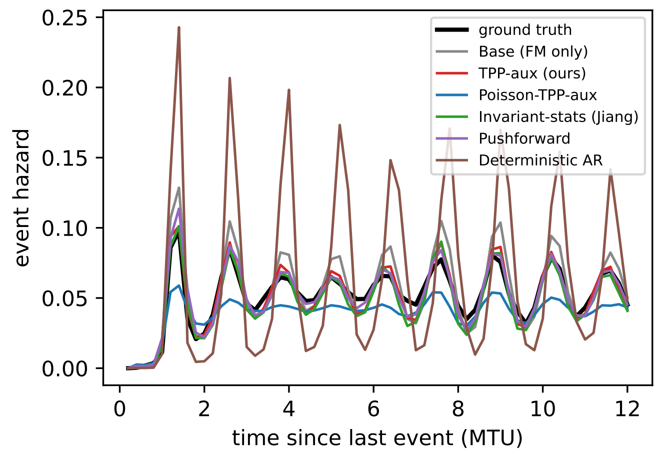

Figure 2: Conditional event hazard vs. time since the last event. Ground truth (black) oscillates at the recurrence period. The base surrogate over-amplifies the oscillation; TPP-aux (red) matches it; the rate-matched Poisson prior (blue) flattens it — the content-specificity dissociation in a single curve.

estimators [2] would remove that bias at higher variance — we found the biased low-variance path sufficient. (v) In noise-dominated regimes, part of the timing deficit is marginal-reducible and time-marginal objectives recover it; the KS results isolate the purely temporal component that they cannot.

References

[1] Yahya Aalaila, Gerrit Großmann, and Sebastian Vollmer. Seahorse: A unified benchmarking framework for spatiotemporal event modeling. *arXiv preprint arXiv:2607.01022*, 2026.

[2] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In *International Conference on Learning Representations*, 2024.

[3] Johannes Brandstetter, Daniel Worrall, and Max Welling. Message passing neural PDE solvers. In *International Conference on Learning Representations*, 2022.

[4] Dibyajyoti Chakraborty, Arvind T. Mohan, and Romit Maulik. Binned spectral power loss for improved prediction of chaotic systems. *Journal of Computational Physics*, 558:114866, 2026. doi: 10.1016/j.jcp.2026.114866.

[5] Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes: Volume I: Elementary Theory and Methods.* Springer, 2003.

[6] Ruoxi Jiang, Peter Y. Lu, Elena Orlova, and Rebecca Willett. Training neural operators to preserve invariant measures of chaotic attractors. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[7] Aly-Khan Kassam and Lloyd N. Trefethen. Fourth-order time-stepping for stiff PDEs. *SIAM Journal on Scientific Computing*, 26(4):1214–1233, 2005.

[8] Dmitrii Kochkov, Janni Yuval, Ian Langmore, Peter Norgaard, Jamie Smith, Griffin Mooers, Milan Klöwer, James Lottes, Stephan Rasp, Peter Düben, Sam Hatfield, Peter Battaglia, Alvaro Sanchez-Gonzalez, Matthew Willson, Michael P. Brenner, and Stephan Hoyer. Neural general circulation models for weather and climate. *Nature*, 632:1060–1066, 2024. doi: 10.1038/s41586-024-07744-y.

[9] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.

[10] Qi Liu, Laure Zanna, and Joan Bruna. Towards infinitely long neural simulations: Self-refining neural surrogate models for dynamical systems. *arXiv preprint arXiv:2603.17750*, 2026.

[11] Edward N. Lorenz. Predictability: A problem partly solved. In *Proc. Seminar on Predictability*, volume 1. ECMWF, 1996.

[12] Weijian Luo, Tianyang Hu, Shifeng Zhang, Jiacheng Sun, Zhenguo Li, and Zhihua Zhang. Diff-Instruct: A universal approach for transferring knowledge from pre-trained diffusion models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[13] Peter Manshausen, Noah Brenowitz, Jan Berner, Karthik Kashinath, and Michael Pritchard. Towards accurate extreme event likelihoods from diffusion model climate emulators. *arXiv preprint arXiv:2605.03802*, 2026.

[14] Hongyuan Mei and Jason Eisner. The neural Hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[15] André Melo, Luis Santiago, and Peter Lu. Learning to emulate chaos: Adversarial optimal transport regularization. *arXiv preprint arXiv:2604.21097*, 2026.

[16] Fengxiang Nie and Yasuhiro Suzuki. Jacobian-adaptive weighting for stability: Enhancing long-term rollout of neural partial differential equation solvers via spatially-adaptive regularization. *arXiv preprint arXiv:2603.05538*, 2026.

[17] Chris Pedersen, Laure Zanna, and Joan Bruna. Thermalizer: Stable autoregressive neural emulation of spatiotemporal chaos. In *International Conference on Machine Learning*, 2025. arXiv:2503.18731.

[18] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637:84–90, 2025. doi: 10.1038/s41586-024-08252-9.

[19] Kornelius Raeth and Nicole Ludwig. Decision-aware training for sample-based generative models. *arXiv preprint arXiv:2607.01171*, 2026.

[20] Yair Schiff, Zhong Yi Wan, Jeffrey B. Parker, Stephan Hoyer, Volodymyr Kuleshov, Fei Sha, and Leonardo Zepeda-Núñez. DySLIM: Dynamics stable learning by invariant measure for chaotic systems. *arXiv preprint arXiv:2402.04467*, 2024. ICML 2024.

[21] Oleksandr Shchur, Marin Biloš, and Stephan Günnemann. Intensity-free learning of temporal point processes. In *International Conference on Learning Representations*, 2020.

[22] Stamatis Stamatelopoulos and Themistoklis P. Sapsis. Can diffusion models capture extreme event statistics? *Computer Methods in Applied Mechanics and Engineering*, 435:117589, 2025. doi: 10.1016/j.cma.2024.117589.

[23] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. ProlificDreamer: High-fidelity and diverse text-to-3D generation with variational score distillation. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[24] Oliver Watt-Meyer, Brian Henn, Jeremy McGibbon, Spencer K. Clark, Anna Kwa, W. Andre Perkins, Elynn Wu, Lucas Harris, and Christopher S. Bretherton. ACE2: accurately learning subseasonal to decadal atmospheric variability and forced responses. *npj Climate and Atmospheric Science*, 8:205, 2025. doi: 10.1038/s41612-025-01090-0.

[25] Daniel S. Wilks. Effects of stochastic parametrizations in the Lorenz '96 system. *Quarterly Journal of the Royal Meteorological Society*, 131(606):389–407, 2005.

A Experimental details

Data. Two-scale Lorenz-96: RK4 at dt=0.002 (validated against 0.001: climatological mean/std/q_{98} agree to $< 0.3\%$), observed every Δt=0.2 MTU after 20 MTU burn-in. Train: 64 trajectories × 160 MTU (51,136 transitions); val: 16 × 160; evaluation references: 64 × 400 MTU plus 8 × 1500 MTU held-out clean runs. Observation noise $\mathcal{N}(0, (0.3\sigma_x)^2)$ applied to train and val only. KS: ETDRK4 [7] at dt=0.25, L=22, N=64, observed every 1 tu; splits proportional.

Event definitions. L96: upcrossings of $s_k = -x_k$ above the pooled clean 95th percentile ($u = 3.082$); soft temperature $\tau = 0.025\,\sigma_x = 0.088$ (soft/hard rate bias +4.9%, 80% of soft mass on hard-event steps; 2τ: +17.5%/62%; 4τ: +53%/38%). KS: upcrossings of u_i above its 95th percentile. Ground-truth structure at the emulator step: L96 troughs sub-Poisson (IET CV 0.78; lag-1 IET correlation ≈ 0; $\mathrm{Fano}_2/\mathrm{Fano}_{20} = 0.71/0.62$); KS crests super-Poisson (CV 1.58, $\mathrm{Fano}_{50} \approx 1.11$).

Models and training. Surrogate backbone: stem + 4 circular-conv residual blocks (width 96, kernel 5, GroupNorm, FiLM conditioning on the flow time), zero-init head; ≈0.66M parameters; residual parameterization $g = (x^n_{t+1} - x^n_t)/\sigma_d$. Base training: 20k steps, AdamW, lr 3×10^{-4}, batch 512, EMA 0.999. Fine-tuning (all conditions): 800 steps, lr 10^{-4}, EMA 0.995, $\lambda_{\mathrm{aux}} = 10$ (TPP-family), $\lambda_{\mathrm{marg}} = 1$; selected on a development seed excluded from all reported statistics; the Poisson control uses the tuned TPP settings unchanged. Sampler: 6 Euler steps. TPP: 1-layer GRU (hidden 64) + 2-layer head, softplus intensity; fit 3k steps on catalog streams (windows 512, 64-step warmup masked).

Auxiliary rollouts. 16 rollouts × 100 steps per aux application (every 2nd optimizer step), initialized at training states, prefixed with 100 steps of catalog event history (likelihood masked over the prefix). Gradients truncated to 25-step windows: states and noises are recorded in a gradient-free rollout, then all windows re-forward in parallel with gradients from their recorded starts — identical values and gradient semantics to truncated BPTT at a fraction of the sequential cost. The gradient-free rollout is executed as a captured CUDA graph. Self-TPP: initialized from the Poissonized TPP, warm-started on 40 init-model rollout batches, then 5 Adam steps (lr 3×10^{-3}) per aux application on the surrogate's hard rollout events. Rollouts with $\max_k |x^n| > 8$ are censored from the aux loss (none occurred in reported runs).

Evaluation. 64 rollouts × 4000 steps from held-out clean initial conditions per checkpoint; censoring at 8σ (no rollout was censored in any reported condition). Return periods computed as pooled stream-time per upcrossing at clean thresholds $q \in \{0.90, 0.95, 0.98, 0.99, 0.995\}$. CRPS: 256

initial conditions, 16-member ensembles. Hazard curves: pooled empirical discrete hazard, bins with ≥ 50 at-risk intervals.

Compute. All experiments on one RTX 3080 (10 GB). Base training $\approx$9 min; TPP-family fine-tune $\approx$20 min; the full L96 confirmatory grid $\approx$8 h.